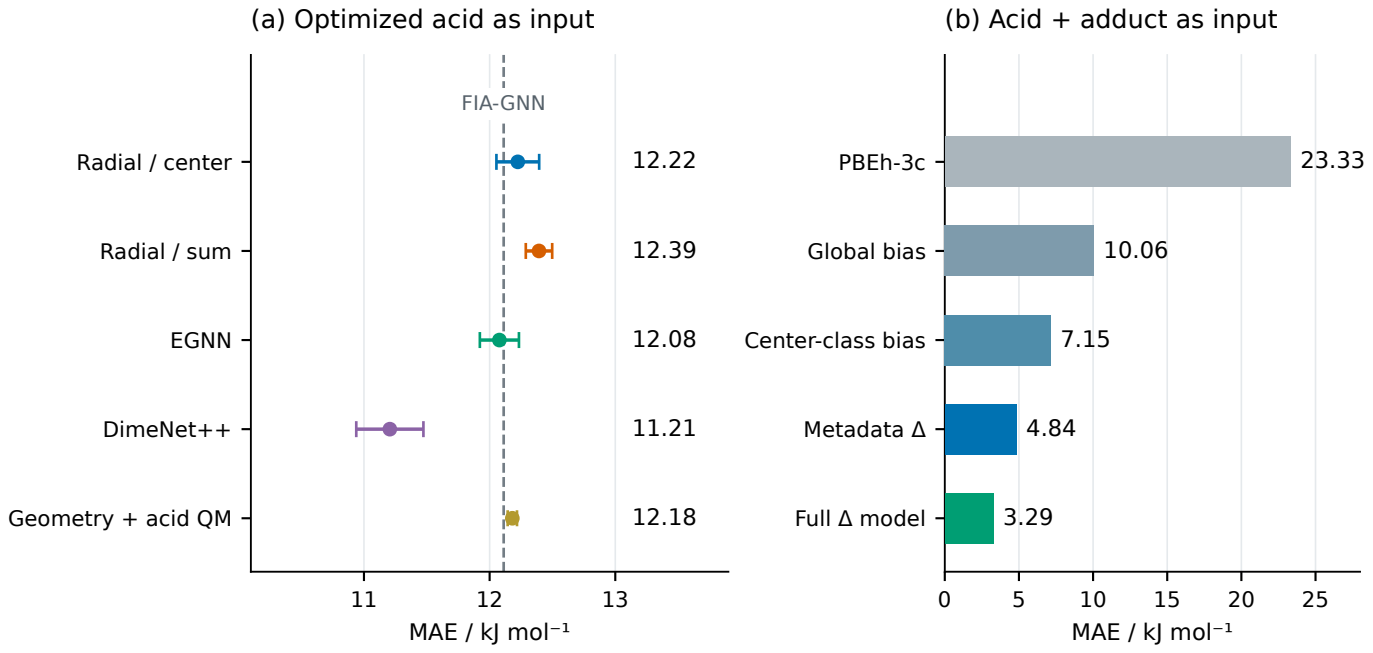
(a) Optimized acid as input
FIA-GNN
Radial / center
Radial / sum
EGNN
DimeNet++
Geometry + acid QM
12.22
12.39
12.08
11.21
12.18
11
12
13
MAE / kJ mol⁻¹
(b) Acid + adduct as input
PBEh-3c
Global bias
Center-class bias
Metadata Δ
Full Δ model
23.33
10.06
7.15
4.84
3.29
0
5
10
15
20
25
MAE / kJ mol⁻¹